Martingale Innovations from Contractive Recurrent Networks and Dimension-Robust Change-Point Detection

YUAN-CHIN IVAN CHANG[a]

Institute of Statistical Science, Academia Sinica, 128 Academia Road, Section 2, Nankang, Taipei 115, Taiwan. [a]ycchang@stat.sinica.edu.tw

Recurrent neural networks (RNNs) are widely used for sequential data, yet a basic question remains open: what does the hidden state of a trained RNN actually encode about its input sequence? A related gap affects change-point detection: standard cumulative sum (CUSUM) procedures assume independent or parametrically specified input, and their false-alarm guarantees break down when the serial dependence is unknown or the observation dimension is large.

This paper addresses both gaps. We establish that the hidden state of a contractive RNN decomposes, at every step, into a predictable component and an innovation that satisfies the martingale-difference property; under mild conditions the accumulated innovations satisfy a functional central limit theorem. Building on this, we introduce the Innovation CUSUM, which applies a CUSUM statistic to LSTM prediction innovations. Because these innovations are approximately free of serial dependence by construction, the detector's false-alarm rate remains well-controlled as the observation dimension grows, without requiring knowledge of the dependence structure.

Eight empirical studies on synthetic and real data support the theory. Applications to the Nile River discharge series and US equity sector returns around the 2020 COVID crash are consistent with the theoretical predictions.

Keywords: martingale decomposition; recurrent neural network; LSTM; change-point detection; CUSUM; average run length; dimension stability; locally stationary processes

1. Introduction

Sequential data arise throughout science and engineering: physiological signals in intensive care, infectious disease incidence, machinery sensor streams. In each setting the central question is the same — how much of the past is relevant to the present? Classical time-series models answer this parametrically. Autoregressive models fix the lag order; hidden Markov models encode memory through a finite state space; Kalman filters propagate a Gaussian sufficient statistic (Box et al., 2015, Hamilton, 1994). Each framework carries an explicit probabilistic account of what is remembered.

Deep sequential architectures aspire instead to learn the relevant memory horizon from data. Recurrent neural networks (RNNs) (Rumelhart, Hinton and Williams, 1986), long short-term memory networks (LSTMs) (Hochreiter and Schmidhuber, 1997), gated recurrent units (GRUs) (Cho et al., 2014), and selective state-space models (SSMs) such as Mamba (Gu and Dao, 2023) implement increasingly sophisticated gating mechanisms for this purpose. Yet a fundamental asymmetry persists: the empirical performance of these architectures is well documented, while the probabilistic account of what they retain is almost entirely absent. What does a trained LSTM encode in its cell state? What does the forget gate actually discard?

The probabilistic object formalizing what survives after any finite prefix is the *tail σ-algebra* $\mathcal{T}_\infty = \bigcap_{n\geq 1} \sigma(X_n, X_{n+1}, \ldots)$; Theorem 2.5 makes this framing rigorous for contractive RNNs.

The natural tool is reverse martingale theory (Doob, 1953, Williams, 1991): sequences adapted to a decreasing filtration converge automatically in both almost-sure and L^1 senses, the rigorous backbone of our proofs.

This paper establishes four results connecting this classical theory to recurrent architectures and change-point detection. The first result shows that, under a contraction condition on the recurrent weight matrix (Assumption 2.4), the hidden state splits at every step into a component fully determined by the past and a genuinely new component that carries only the fresh information in the current observation. This new component stabilizes geometrically and has a non-degenerate variance structure (Theorem 2.5); the stationary causal solution used as background is established in Chang (2026). The second result shows that the accumulated new-information components, suitably normalized, converge in distribution to a Brownian motion, providing the asymptotic theory needed to calibrate inference procedures built on these innovations (Corollary 2.8).

The third and fourth results concern change-point detection. We prove that a cumulative sum (CUSUM) detector built on LSTM prediction innovations — which inherit the new-information property above — has a false-alarm rate that is exponentially controlled in the detection threshold, and that this guarantee remains stable as the observation dimension grows (Theorem 3.2 and Corollary 3.5). Two supporting results underpin the main theorems: a lemma on the equivalence of forward and reverse filtration limits, and a proposition establishing that exact Bayes prediction residuals carry no predictable component (Lemma 2.1 and Proposition 2.9).

Existing work on recurrent architectures addresses stability and expressibility (Miller and Hardt, 2019, Miyato et al., 2018, Gonon and Ortega, 2020) rather than the informational content of the hidden state. The neural tangent kernel (Jacot, Gabriel and Hongler, 2018) and information bottleneck (Tishby and Zaslavsky, 2015, Saxe et al., 2019) frameworks apply to feedforward networks or to compression at a fixed time step, not to sequential memory. Structured state-space models and Mamba (Gu, Goel and Ré, 2022, Gu and Dao, 2023) have been studied mainly for computational efficiency (Orvieto et al., 2023). On the detection side, standard CUSUM (Page, 1954, Lorden, 1971), offline segmentation (Killick, Fearnhead and Eckley, 2012), and Bayesian online methods (Adams and MacKay, 2007) are calibrated for independent or fully specified parametric input and lose their guarantees under unknown serial dependence or growing dimension. Reverse martingale convergence is classical (Doob, 1953, Neveu, 1975, Hall and Heyde, 1980) but its application to learned sequential architectures appears unexplored. The non-stationary extension uses the locally stationary framework of Dahlhaus (1997).

The theoretical framework and main results appear in Sections 2–3, followed by eight empirical studies in Section 4. Background definitions and all formal proofs are given in the Appendices and Supplementary Material.

2. Methodology

We work on a complete probability space $(\Omega, \mathcal{A}, \mathbb{P})$. Let $(X_t)_{t\geq 1}$ be a strictly stationary ergodic process with natural filtration $\mathcal{F}_t = \sigma(X_1, \ldots, X_t)$ and tail σ-algebra $\mathcal{T}_\infty = \bigcap_{t\geq 1} \sigma(X_t, X_{t+1}, \ldots)$. An RNN processes the sequence via $h_t = \phi(W_h h_{t-1} + W_x x_t + b_h)$ and an LSTM augments this with a gated cell state; both architectures are defined precisely in Section A of the Supplementary Material. Throughout Sections 2–3 we use p to denote the hidden-state dimension of the RNN and d_{in} for the input dimension. In Section 4 the symbol d refers to the observation (input) dimension of the multivariate DGP, following the change-point literature convention. Throughout, we assume the effective recurrent Lipschitz constant satisfies $L_\phi \|W_h\|_{\text{op}} < 1$ (operator-norm contractivity; see Assumption 2.4 below), which can be encouraged in practice by spectral normalization (Miyato et al., 2018). For a fixed target

$Z \in L^2$, we use the *tail-excess information*

$$\mathcal{I}_t^{\mathrm{tail}}(Z) = \left\| \mathbb{E}[Z \mid \mathcal{F}_t] - \mathbb{E}[Z \mid \mathcal{T}_\infty] \right\|_{L^2},$$

as a diagnostic of the gap between finite-history prediction and the tail projection. This quantity is not asserted to vanish for every fixed target; its limit is zero only when the forward limit agrees with the tail projection. For one-step prediction, we instead use the innovation $e_{t+1} = Y_{t+1} - \mathbb{E}[Y_{t+1} \mid \mathcal{F}_t]$, whose martingale-difference sequence (MDS) property is stated separately in Proposition 2.9. A process (X_t) is *absolutely regular* (β-mixing) if its mixing coefficients $\beta(k) = \sup_t \mathbb{E}\{\sup_{B \in \sigma(X_{t+k}, X_{t+k+1}, \dots)} |\mathbb{P}(B \mid \mathcal{F}_t) - \mathbb{P}(B)|\}$ tend to zero. Such assumptions are used only for empirical motivation and local approximation, not to claim that (h_t) alone is a Markov chain. Full definitions, the reverse martingale convergence theorem, and the i.i.d. Hewitt–Savage zero-one law are collected in Section A of the Supplementary Material.

2.1. Equivalence of Forward and Tail Limits

The first result resolves the apparent tension between the *forward* perspective (the hidden state is adapted to the increasing filtration $\mathcal{F}_t$) and the *reverse* perspective (selective memory is characterized by the decreasing filtration $\mathcal{G}_n$).

Lemma 2.1 (Equivalence of forward and tail limits). *Let $(X_t)_{t\geq 1}$ be a stationary ergodic sequence on $(\Omega, \mathcal{A}, \mathbb{P})$, and let $\theta : \Omega \to \mathbb{R}$ be bounded and tail-measurable, i.e. $\theta \in L^\infty(\Omega, \mathcal{T}_\infty, \mathbb{P})$. Define the natural filtration $\mathcal{F}_t = \sigma(X_1, \dots, X_t)$ and the reverse filtration $\mathcal{G}_n = \sigma(X_n, X_{n+1}, \dots)$, so that $\mathcal{T}_\infty = \bigcap_{n\geq 1} \mathcal{G}_n$. Then:*

(a) $\mathbb{E}[\theta \mid \mathcal{F}_t] \xrightarrow{a.s.\text{ and } L^1} \theta$ *as $t \to \infty$.*

(b) $\mathbb{E}[\theta \mid \mathcal{G}_n] \xrightarrow{a.s.\text{ and } L^1} \theta$ *as $n \to \infty$.*

(c) *Both limits equal θ almost surely; consequently,* $\lim_{t\to\infty} \mathbb{E}[\theta \mid \mathcal{F}_t] = \lim_{n\to\infty} \mathbb{E}[\theta \mid \mathcal{G}_n]$ *a.s.*

For a non-tail-measurable $\theta \in L^1(\Omega, \mathcal{A}, \mathbb{P})$:

(d) $\mathbb{E}[\theta \mid \mathcal{F}_t] \xrightarrow{a.s.} \mathbb{E}[\theta \mid \mathcal{F}_\infty^+]$*, which may be strictly larger (in σ-algebra terms) than $\mathcal{T}_\infty$.*

(e) $\mathbb{E}[\theta \mid \mathcal{G}_n] \xrightarrow{a.s.} \mathbb{E}[\theta \mid \mathcal{T}_\infty]$*, the projection of θ onto $\mathcal{T}_\infty$, which extracts only the structurally persistent component.*

Proof. See Section B of the Supplementary Material. □

Remark 2.2 (Interpretation for sequential learning). Lemma 2.1 resolves only the conditional-expectation part of the forward/reverse tension: if the target is already tail-measurable, then conditioning on the increasing natural filtration and conditioning on the decreasing tail filtration identify the same limit. It does *not* by itself imply that an RNN hidden state converges to a tail projection. The hidden state is a stochastic recurrence driven by the observations, and its long-run behavior must be analyzed separately, as in Theorem 2.5.

Remark 2.3 (Non-tail-measurable targets). Parts (d) and (e) show that when the target θ is not tail-measurable, the forward and reverse limits *differ*: the forward limit $\mathbb{E}[\theta \mid \mathcal{F}_\infty^+]$ may incorporate information from the finite initial segment, while the reverse limit $\mathbb{E}[\theta \mid \mathcal{T}_\infty]$ discards it. This formalizes *selective forgetting*: the reverse martingale extracts only the structurally persistent (tail-measurable) component of any target.

Assumption 2.4 (Effective operator-norm contractivity). Throughout the paper we assume $L_\phi \|W_h\|_{\mathrm{op}} < 1$, where L_ϕ is the Lipschitz constant of ϕ and $\|W_h\|_{\mathrm{op}} = \sigma_{\max}(W_h)$. Spectral normalization (Miyato et al., 2018) can encourage this condition in practice.

2.2. Martingale Decomposition of RNN Hidden States

Throughout this subsection we work under Assumption 2.4 ($\rho := L_\phi \|W_h\|_{\mathrm{op}} < 1$). The existence and uniqueness of the stationary causal solution $(H_t^*)_{t\in\mathbb{Z}}$ satisfying $H_t^* = \phi(W_h H_{t-1}^* + W_x X_t + b_h)$, $H_t^* \in \sigma(X_t, X_{t-1}, \ldots)$, and the geometric coupling bound $\left\|h_t - H_t^*\right\|_{L^2} \le \rho^t \left\|h_0 - H_0^*\right\|_{L^2}$ are established in Chang (2026); a self-contained proof is also given in the Supplementary Material. We write $\Delta_0 := \left\|h_0 - H_0^*\right\|_{L^2}$, $D_t^* := \mathbb{E}[H_t^* \mid \mathcal{F}_{t-1}]$ (stationary predictable component), and $M_t^* := H_t^* - D_t^*$ (stationary MDS innovation). The stationary causal solution is generally a *random* process; it should not be identified with a deterministic mean-field fixed point or a tail projection.

Theorem 2.5 (MDS Decomposition with Stabilization Rate). *Let $(X_t)_{t\ge1}$ be a strictly stationary ergodic process with $\mathbb{E}\|X_t\|^2 < \infty$, and let $(h_t)_{t\ge0}$ be the RNN hidden-state sequence with square-integrable initial state h_0. Set $\rho = L_\phi \|W_h\|_{\mathrm{op}} < 1$ and $\Delta_0 = \left\|h_0 - H_0^*\right\|_{L^2}$. Then, for each $t \ge 2$:*

(i) *(L^2**-boundedness.**) (h_t) is uniformly L^2-bounded:*

$$\sup_t \|h_t\|_{L^2} \le \|h_0\|_{L^2} + \frac{L_\phi\left(\|W_x\|_{\mathrm{op}} \|X_1\|_{L^2} + \|b_h\|\right)}{1-\rho} =: C_0 < \infty.$$

(ii) ***(MDS decomposition.)*** *The Doob decomposition $h_t = D_t + M_t$ holds, where $D_t := \mathbb{E}[h_t \mid \mathcal{F}_{t-1}]$ is $\mathcal{F}_{t-1}$-measurable (predictable) and $M_t := h_t - D_t$ satisfies $\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] = 0$ a.s. The pair $(M_t, \mathcal{F}_t)_{t\ge2}$ is a martingale difference sequence.*

(iii) ***(Innovation L^2 bound.)*** *Writing* $\mathrm{Var}_2(X_t) := \mathbb{E}\|X_t - \mathbb{E}X_t\|^2$,

$$\mathbb{E}\|M_t\|^2 \le L_\phi^2 \|W_x\|_{\mathrm{op}}^2 \,\mathrm{Var}_2(X_t) =: \sigma_M^2. \tag{1}$$

(iv) ***(Innovation stabilization rate.)*** [New.] *The MDS innovations converge to their stationary counterparts at a geometric rate:*

$$\left\|M_t - M_t^*\right\|_{L^2} \le 2\rho^t \Delta_0. \tag{2}$$

(v) ***(Conditional variance decomposition and stationary innovation covariance.)*** [New.] *The law-of-total-variance decomposition holds exactly:*

$$\mathbb{E}\|h_t - \mathbb{E}h_t\|^2 = \mathbb{E}\|D_t - \mathbb{E}h_t\|^2 + \mathbb{E}\|M_t\|^2. \tag{3}$$

In the stationary regime the innovation covariance matrix

$$\Sigma_M^* := \mathbb{E}\left[M_t^* (M_t^*)^\top\right] \tag{4}$$

is well-defined, independent of t, positive semi-definite, and satisfies $\Sigma_M^* \preceq \sigma_M^2 I_p$. Moreover $\lambda_{\min}(\Sigma_M^*) > 0$ under the following nondegeneracy condition: for every nonzero $v \in \mathbb{R}^p$,

$$\mathrm{Var}\!\left(v^\top \phi(W_h H_{t-1}^* + W_x X_t + b_h) \,\middle|\, \mathcal{F}_{t-1}\right) > 0 \quad a.s. \tag{5}$$

A sufficient condition for (5) is that X_t is not entirely predictable from $\mathcal{F}_{t-1}$ and W_x has full row rank $\operatorname{rank}(W_x) = p$*; however, this sufficient condition may fail for saturating nonlinearities or when $d_{\text{in}} < p$ (see Remark 2.7).*

Proof. See Section B of the Supplementary Material. □

Remark 2.6 (What is new). Parts (i)–(iii) parallel the earlier Proposition 2.9 argument but now apply to the hidden state itself rather than the prediction target. **Parts (iv) and (v) are new.** Part (iv) shows that the transient MDS innovation approaches the stationary innovation at the same geometric rate ρ^t as the hidden state itself, which justifies asymptotic calibration of the Innovation CUSUM from a burn-in period of order $(1-\rho)^{-1}$ steps. Part (v) provides the stationary innovation covariance Σ_M^* needed for the functional central limit theorem (CLT) and for Mahalanobis scoring in the multivariate CUSUM.

Remark 2.7 (Scope of the nondegeneracy condition). The nondegeneracy condition (5) is the cleanest sufficient condition for $\lambda_{\min}(\Sigma_M^*) > 0$: it directly requires that the innovation M_t^* has non-zero variance in every direction. The full row rank condition $\operatorname{rank}(W_x) = p$ is a natural structural sufficient condition for (5) under linear activation, but it can fail to imply (5) for saturating nonlinearities (tanh, sigmoid) when the conditional support of X_t lies in a region where the activation is nearly flat. In the common regime $d_{\text{in}} < p$ (e.g. scalar input with $p = 32$ hidden units), W_x cannot have full row rank, and Σ_M^* is at most rank-d_{in}. In practice one should verify positive definiteness from a pilot run. The functional CLT (Corollary 2.8) requires positive definiteness as a stated hypothesis rather than deriving it from W_x.

Corollary 2.8 (Functional CLT for MDS innovations). *In addition to the hypotheses of Theorem 2.5, assume $\mathbb{E}\|X_t\|^{2+\delta} < \infty$ for some $\delta > 0$ and that Σ_M^* is positive definite. Define the partial-sum process $\mathbf{S}_T(u) := T^{-1/2}\sum_{t=1}^{\lfloor Tu\rfloor} M_t$, $u \in [0,1]$. Then*

$$\mathbf{S}_T(\cdot) \xrightarrow{d} (\Sigma_M^*)^{1/2}\, W(\cdot) \qquad \text{in } D\big([0,1],\mathbb{R}^p\big), \tag{6}$$

where W is a standard p-dimensional Brownian motion and $D([0,1],\mathbb{R}^p)$ denotes the Skorokhod space of right-continuous left-limited functions (the natural domain for the step-function partial-sum process). The transient-correction process $\mathbf{T}_T(u) := T^{-1/2}\sum_{t=1}^{\lfloor Tu\rfloor}(M_t - M_t^*)$ satisfies $\sup_{u\in[0,1]}\|\mathbf{T}_T(u)\| \to 0$ in probability, with explicit rate

$$\mathbb{E}\left[\sup_{u\in[0,1]}\|\mathbf{T}_T(u)\|\right] \le \frac{2\Delta_0\rho}{\sqrt{T}(1-\rho)} \to 0$$

(see proof for the sup-norm bound and the functional Slutsky argument), so (6) holds for (M_t) and (M_t^*) simultaneously.

Proof. See Section B of the Supplementary Material. □

Proposition 2.9 (Bayes prediction residuals are martingale differences). *Let Y_{t+1} be a square-integrable, $\mathcal{F}_{t+1}$-measurable prediction target and let $m_t = \mathbb{E}[Y_{t+1} \mid \mathcal{F}_t]$. Then*

$$e_{t+1} = Y_{t+1} - m_t$$

is $\mathcal{F}_{t+1}$-measurable, satisfies $\mathbb{E}[e_{t+1} \mid \mathcal{F}_t] = 0$, and hence is a martingale difference relative to $(\mathcal{F}_t)$. If an $\mathcal{F}_t$-measurable estimator $\hat{m}_t$ obeys $\sup_t \|\hat{m}_t - m_t\|_{L^2} \le \eta$, then the fitted residuals $\hat{e}_{t+1} = Y_{t+1} - \hat{m}_t$ are approximate martingale differences in the sense that $\sup_t \|\mathbb{E}[\hat{e}_{t+1} \mid \mathcal{F}_t]\|_{L^2} \le \eta$.

Proof. See Section B of the Supplementary Material. □

2.3. Extension to Non-Stationary Processes

The results in Sections 2.1–2.2 require stationarity. We now describe how they extend to the locally stationary setting of Dahlhaus (1997).

Definition 2.10 (Locally stationary process; Dahlhaus 1997). A triangular array $(X_{t,T})_{1\le t\le T}$ is *locally stationary* with transfer function $A^0(u,\cdot)$ if there exists a representation

$$X_{t,T} = \int_{-\pi}^{\pi} e^{i\lambda t} A^0(t/T,\lambda)\, dZ(\lambda),$$

where $Z(\lambda)$ is a complex-valued orthogonal increment process and $A^0(u,\lambda)$ is a smooth function of the rescaled time $u = t/T \in [0,1]$.

Corollary 2.11 (Approximate local stationary decomposition). *Under the locally stationary model of Definition 2.10, assume that $A^0(u,\lambda)$ is Lipschitz in u uniformly in λ (Dahlhaus, 1997, Assumption 2). Fix $u \in (0,1)$ and let $X_t^{(u)}$ denote the stationary approximating process with transfer function $A^0(u,\cdot)$. Let $h_{t,T}$ and $h_t^{(u)}$ be the corresponding RNN states driven by $X_{t,T}$ and $X_t^{(u)}$, respectively, under the same contractive recurrence and a common square-integrable initialization at the left edge of the local block. Let $B_T(u)$ be any block of at most m_T consecutive time indices satisfying $|t/T-u| \le b_T$ for all $t \in B_T(u)$. If $b_T \downarrow 0$, $m_T \to \infty$, and $m_T/T \to 0$, then*

$$\sup_{t\in B_T(u)} \left\| h_{t,T} - h_t^{(u)} \right\|_{L^2} = O\!\left(b_T + \frac{m_T}{T}\right) + o(1). \tag{7}$$

The stationary approximation $h_t^{(u)}$ has the decomposition of Theorem 2.5, with all bounds evaluated under the local law at rescaled time u and spectral density $f(u,\lambda) = |A^0(u,\lambda)|^2$. Thus the martingale-difference decomposition is local and approximate rather than an almost-sure convergence statement to a literal "local tail" limit of the finite triangular array.

Proof. See Section B of the Supplementary Material. □

3. Innovation CUSUM: In-Control False-Alarm Theory

We now give a rigorous in-control performance guarantee for the Innovation CUSUM, the change-point detector that applies a Page CUSUM to LSTM prediction innovations. By Proposition 2.9, exact Bayes residuals form an MDS; by Corollary 2.8, the LSTM-fitted residuals are approximate MDS whose partial sums satisfy a functional CLT. The key consequence for the alarm rate follows from a sub-Gaussian exponential supermartingale argument.

3.1. Setup and Assumptions

The one-sided Page CUSUM (Page, 1954) is

$$S_0 = 0, \qquad S_t = (S_{t-1} + \hat{e}_t - k)^+, \qquad \tau_h = \inf\{t \ge 1 : S_t \ge h\}, \tag{8}$$

where $k > 0$ is a reference (slack) value and $h > 0$ is the alarm threshold.

Assumption 3.1 (Innovation CUSUM operating conditions).

(B1) **(Approximation quality.)** The LSTM predictor $\hat{m}_t$ satisfies $\eta := \sup_t \|\hat{m}_t - m_t\|_{L^2} < \infty$, where $m_t = \mathbb{E}[Y_{t+1} \mid \mathcal{F}_t]$ is the Bayes predictor.

(B2) **(Conditional sub-Gaussianity.)** In the in-control regime there exists $\sigma > 0$ such that $\mathbb{E}[\exp\{\lambda(\hat{e}_t - \mathbb{E}[\hat{e}_t \mid \mathcal{F}_{t-1}])\} \mid \mathcal{F}_{t-1}] \le e^{\lambda^2\sigma^2/2}$ for all $\lambda \in \mathbb{R}$. This holds when $\hat{e}_t \in [-B, B]$ a.s. with $\sigma = B$.

(B3) **(Reference value exceeds bias.)** $k > \eta$; write $\kappa := k - \eta > 0$ for the *net drift margin*.

(B4) **(In-control mean condition.)** $\mathbb{E}[\hat{e}_t \mid \mathcal{F}_{t-1}] \le \eta$ a.s. in the in-control regime, so that $\mathbb{E}[\hat{e}_t - k \mid \mathcal{F}_{t-1}] \le -\kappa < 0$.

3.2. Main Result: Exponential ARL Lower Bound

Theorem 3.2 (In-control ARL lower bound for the Innovation CUSUM). *Under Assumption 3.1, the in-control average run length (ARL$_0$; here $\mathbb{E}_0[\cdot]$ denotes expectation under the in-control distribution, i.e. prior to any change-point) satisfies*

$$\mathbb{E}_0[\tau_h] \;\ge\; \frac{1}{2\sqrt{2}}\exp\left(\frac{\kappa h}{2\sigma^2}\right), \tag{9}$$

where $\kappa = k - \eta$ is the net drift margin and σ^2 the sub-Gaussianity parameter. Equivalently, to achieve an in-control ARL of at least L it suffices to set $h \ge (2\sigma^2/\kappa)\ln(2\sqrt{2}\,L)$.

Proof. See Section B of the Supplementary Material. □

Remark 3.3 (Practical interpretation). The ARL grows exponentially in h with rate $\kappa/(2\sigma^2)$, where $\kappa = k - \eta$ is the net margin between the reference value and the LSTM approximation bias. The bound degrades gracefully as η increases, becoming vacuous only when $\eta \to k$. The threshold formula $h \ge (2\sigma^2/\kappa)\ln(2\sqrt{2}L)$ enables principled calibration from η and σ^2 without distributional knowledge. The exponent $\kappa/(2\sigma^2)$ is conservative by a factor of 4 relative to the classical i.i.d. formula $2\kappa/\sigma^2$; closing this gap via optional stopping theory is left for future work.

Remark 3.4 (One-sided vs. two-sided CUSUM). Theorem 3.2 is stated for the one-sided Page CUSUM $\tau_h = \inf\{t : S_t \ge h\}$ in equation (8). The empirical studies use a two-sided version that monitors both upward and downward shifts simultaneously, with alarm time $\tau_h^{\pm} = \min(\tau_h^+, \tau_h^-)$ where $\tau_h^{\pm}$ track positive and negative cumulative sums respectively. By a union bound, the in-control ARL of the two-sided procedure satisfies $\mathbb{E}_0[\tau_h^{\pm}] \ge \mathbb{E}_0[\tau_h]/2$, so the lower bound (9) extends to the two-sided case with an additional factor of $1/2$. The qualitative exponential growth in h is preserved.

3.3. Dimension Stability

Corollary 3.5 (Dimension-stable ARL). *Let $\hat{e}_t \in \mathbb{R}^d$ be the multivariate LSTM innovation and define the aggregated Mahalanobis score*

$$\mathcal{Z}_t := \frac{1}{d}(\hat{e}_t - \mu_0)^\top \Sigma_0^{-1}(\hat{e}_t - \mu_0) - 1,$$

where $\mu_0 = \mathbb{E}[\hat{e}_t]$ and $\Sigma_0 = \mathrm{Var}(\hat{e}_t)$. Suppose Assumption 3.1 holds for the innovation vector with $\eta_d = O(1)$ (LSTM approximation quality bounded uniformly in d) and $\sigma_d = O(1)$ as $d \to \infty$, and that the centered innovation $u_t := \hat{e}_t - \mu_0$ satisfies $\mathbb{E}[u_t u_t^\top \mid \mathcal{F}_{t-1}] \preceq \Sigma_0$ a.s. (conditional homoskedasticity). Assume further that the in-control innovation covariance is uniformly non-degenerate: $\lambda_{\min}(\Sigma_0) \geq c_0 > 0$ for all d, so that $\|\Sigma_0^{-1}\|_{\mathrm{op}} = O(1)$ as $d \to \infty$. Then $\eta'_d := 4\|\Sigma_0^{-1}\|_{\mathrm{op}}\eta_d^2/d = O(d^{-1}) \to 0$, the net drift margin $\kappa_d = k_d - \eta'_d \to k > 0$ as $d \to \infty$, and for all sufficiently large d the ARL lower bound (9) holds with $\kappa_d \geq k/2 > 0$.

By contrast, for a raw vector autoregression (VAR(1)) process $X_t = \Phi X_{t-1} + \varepsilon_t$ with coefficient matrix $\Phi \in \mathbb{R}^{d\times d}$, $\|\Phi\|_{\mathrm{op}} = \rho_{\mathrm{op}} > 0$, $\varepsilon_t \overset{\mathrm{iid}}{\sim} (0, I_d)$, $\mu_0 = 0$, stationary covariance $\Sigma_X := \mathrm{Var}(X_t)$, and $\Sigma_0 = I_d$ (identity standardization), the aggregated Mahalanobis score satisfies

$$\mathbb{E}[\mathcal{Z}_t^{\mathrm{raw}} \mid \mathcal{F}_{t-1}] = \frac{1}{d}\,\|\Phi X_{t-1}\|^2 = \frac{1}{d}\,\mathrm{tr}\big(\Phi X_{t-1} X_{t-1}^\top \Phi^\top\big), \tag{10}$$

(the noise-variance term $d^{-1}\mathrm{tr}(I_d) - 1 = 0$ cancels exactly when $\Sigma_0 = I_d$), and this conditional bias need not vanish as $d \to \infty$: for the symmetric case $\Phi = \rho_{\mathrm{op}} I_d$, by the law of large numbers applied to the i.i.d. stationary components of ΦX_{t-1}, $d^{-1}\|\Phi X_{t-1}\|^2 \to \rho_{\mathrm{op}}^2/(1-\rho_{\mathrm{op}}^2) > 0$ a.s.; for general Φ the bias $d^{-1}\mathrm{tr}(\Phi\Sigma_X\Phi^\top)$ is strictly positive for each fixed d whenever $\Phi \neq 0$ and $\Sigma_X \succ 0$, but need not remain bounded away from zero as $d \to \infty$ (e.g. for $\Phi = \mathrm{diag}(\rho_{\mathrm{op}}, 0, \ldots, 0)$, the bias $d^{-1}\mathrm{tr}(\Phi\Sigma_X\Phi^\top) \to 0$). Consequently the net drift margin $\kappa_d^{\mathrm{raw}} = k - \eta_d^{\mathrm{raw}}$ cannot be made positive by a fixed reference value k unless k is recalibrated to exceed the serial-dependence bias η_d^{raw}, which itself depends on Φ and the stationary distribution of X_{t-1} in a way that raw CUSUM calibration does not capture.

Proof. See Section B of the Supplementary Material. □

Remark 3.6 (Connection to Study CUSUM-DIM). Corollary 3.5 provides a formal account of the dimension-scaling pattern observed in Study CUSUM-DIM (Section 4.6): at $d = 10$, the raw and whitened CUSUM ARL_0 falls to 47 and 42 while the Innovation CUSUM rises to 365. Equation (10) shows that the serial-dependence bias η_d^{raw} remains nonzero after the $1/d$ normalization for any $\rho_{\mathrm{op}} > 0$, so a fixed reference value k cannot provide a positive drift margin without explicit recalibration using Φ and Σ_X. This is a calibration requirement, not a universal impossibility: if the reference value k could be perfectly matched to the bias, the raw CUSUM would also be valid. The Innovation CUSUM guarantee holds for all sufficiently large d (specifically once $\eta'_d = 4\|\Sigma_0^{-1}\|_{\mathrm{op}}\eta_d^2/d < k$, which holds for $d \geq d_0$ where d_0 depends on η_d and k).

4. Empirical Studies and Discussion

This section presents eight empirical studies that follow a clear logical progression. Study MDS examines the MDS property (Theorem 2.5) in controlled stationary settings. Study CUSUM-CP then asks whether a CUSUM built on LSTM innovations outperforms classical detectors at a known change-point, empirically testing the prediction of Theorem 3.2. Studies MDS-VAR and KAL extend the MDS diagnostic and prediction benchmarks to multivariate VAR(1) processes, showing that the LSTM advantage is consistent across d. Study CUSUM-DIM examines the dimension-stability of the Innovation CUSUM (Corollary 3.5) across $d \in \{1, 2, 5, 10\}$, finding results consistent with the theoretical prediction. Studies NILE and ETF apply the full pipeline to real datasets — the Nile River and US equity sector returns — bridging theory and practice. Study PELT closes with a direct comparison against the PELT offline detector, answering why a learned-innovation approach is preferable to an established offline change-point method.

Two supporting studies are deferred to the Supplementary Material. Study LS-Diag (Section D.1 of the Supplementary Material) examines locally stationary AR data and finds that a stationary-trained LSTM forget gate systematically mis-adapts to smooth non-stationarity, motivating the non-stationarity correction of Chang (2026). Study J-Surr (Section D.2 of the Supplementary Material) benchmarks four data-driven surrogates for the predictive-decay profile J_t (defined in Section 4.1; naive, forget-gate, block-bootstrap, and a Kullback–Leibler importance estimation procedure (KLIEP)-corrected estimator Sugiyama, Suzuki and Kanamori 2012) and shows that the KLIEP-corrected estimator achieves normalized mean squared error (NMSE) below one under smooth variation but deteriorates under abrupt change; full tables and figures are given there.

All experiments use small-to-moderate sample sizes deliberately, since this paper focuses on theoretical diagnostics rather than application performance. All code is in `code/` (Python 3.9, PyTorch 2.2, and R 4.1), reproducible via `python run_all.py`. Studies MDS and CUSUM-CP use 1000–2000 independent test sequences per configuration; Studies MDS-VAR, KAL, and CUSUM-DIM use 1000 sequences.

4.1. Processes, Benchmarks, and Real Data

Three synthetic data-generating processes and two real datasets are used across all studies. The first is the AR(1) model $X_t = \phi X_{t-1} + \varepsilon_t$, $\varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$, with $\phi \in \{0.3, 0.7, 0.95\}$. For the initialization transient starting at $h_0 = 0$, the transient predictive-decay profile is $J_t = |\phi|^t \cdot \sigma/\sqrt{1-\phi^2}$ (geometric decay).

The second is a two-state hidden Markov model (HMM) with states $s \in \{0, 1\}$, transition matrix $P = \left[\begin{smallmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{smallmatrix}\right]$, and emissions $X_t \mid s_t \sim \mathcal{N}(\pm 1.5, 1)$; the predictive-decay profile J_t is computed numerically via the forward algorithm. The third is a regime-switching autoregression (RS-AR) with AR coefficients $\phi_1 = 0.30$ and $\phi_2 = 0.85$ under the same transition matrix.

The first real series is the Nile River annual discharge at Aswan, 1871–1970 ($T = 100$) (Cobb, 1978). An ordinary least squares (OLS) AR(1) fit gives $\hat{\phi} = 0.980$, $\hat{\sigma} = 166.5$ (10^8m^3/yr). The at-most-one-change (AMOC) change-point estimator (Killick, Fearnhead and Eckley, 2012, Sec. 3.1) recovers the true break at $t = 28$ (year 1898) exactly. Ljung–Box tests on AR(1) residuals reject the MDS null at both lag 5 ($Q = 17.56$, $p = 0.0036$) and lag 10 ($Q = 30.21$, $p = 0.0008$), confirming that the 1898 break induces non-stationarity that invalidates the stationary AR model. The second real dataset is a panel of US equity sector exchange-traded fund (ETF) log-returns described in Section 4.8, examined in a univariate configuration ($d = 1$, using the Financial sector ETF XLF) and a three-dimensional configuration ($d = 3$, XLF together with the Technology sector ETF XLK and the Energy sector ETF XLE), with the 2020-02-19 COVID market peak as the known change-point.

4.2. Study MDS: Stationary Case — Prediction-Residual MDS Diagnostic

We train one `InstrumentedLSTM` (hidden dimension $d = 32$, spectral normalization on W_h, Adam, $N = 1000$ training sequences of length $T = 150$, 40 epochs) on each process and compute one-step-ahead residuals $r_t = X_{t+1} - \hat{y}_t$ on $N_{\text{test}} = 1000$ independent sequences of length $T = 200$. The Ljung–Box statistic (Ljung and Box, 1978) at maximum lag 5 is applied to each sequence; the *MDS pass rate* is the fraction of sequences not rejecting at level 0.05.

Table 1 shows the results. For AR(1) with $\phi \in \{0.3, 0.7\}$ and the HMM, the LSTM pass rate is close to the nominal level $1 - \alpha = 0.95$, consistent with Proposition 2.9 when the fitted predictor is close to the conditional mean. The lower rate for $\phi = 0.95$ (0.386) is expected: a near-unit-root process requires very long sequences to reach its stationary regime; within $T = 200$ steps the LSTM cell state has not fully converged, so the residuals retain residual autocorrelation. The OLS AR(1) baseline confirms that the pattern is process-driven, not a model artifact: OLS residuals for the HMM and RS-AR have notably low pass rates (0.561 and 0.947) because a misspecified linear predictor cannot capture the nonlinear regime structure.

Table 1. Study MDS: Ljung–Box MDS pass rate (fraction of 1000 test sequences not rejecting zero autocorrelation at $\alpha = 0.05$, lag ≤ 5). Bold values are within ±0.05 of the nominal level 0.95.

Process	LSTM pass rate	OLS pass rate	Expected	N_{seq}
AR(1), $\phi = 0.30$	**0.948**	0.969	0.95	1000
AR(1), $\phi = 0.70$	0.905	0.974	0.95	1000
AR(1), $\phi = 0.95$	0.386	0.953	0.95	1000
HMM	**0.953**	0.561	0.95	1000
RS-AR	0.814	0.947	0.95	1000

4.3. Study CUSUM-CP: Innovation CUSUM at a Known Change-Point

Study MDS operated entirely under stationarity. A natural next question is whether an online CUSUM built on LSTM innovations can detect an abrupt structural break reliably, and whether it beats the two classical alternatives. This empirically examines the prediction of Theorem 3.2: if the LSTM residuals approximate a martingale difference sequence in-control, the ARL lower bound should be empirically visible.

For abrupt change-point AR, we generate 2000 sequences with a single abrupt change in the AR coefficient from $\phi_{\text{before}} = 0.30$ to $\phi_{\text{after}} = 0.85$ at $\tau^* = 100$ ($T = 200$). All detectors use Page's two-sided CUSUM (Page, 1954, Lorden, 1971) with the same threshold $h = 5$ and reference value $k = 0.5$, estimated from the first 20% of each sequence. Three methods are compared. M1 (Raw) applies CUSUM to the standardized observations $(X_t - \hat{\mu}_0)/\hat{\sigma}_0$, the standard i.i.d.-calibrated detector. M2 (Whitened) applies CUSUM to AR(1)-whitened residuals $(X_t - \hat{\phi}X_{t-1})/\hat{\sigma}_\varepsilon$, where $\hat{\phi}$ is estimated from the burn-in window; this is the best classical fix, as it explicitly removes lag-1 autocorrelation. M3 (the Innovation CUSUM) applies CUSUM to LSTM one-step-ahead prediction residuals, standardized by in-control moments estimated from a held-out stationary calibration set; *no knowledge of the AR structure is given to the method.*

The key issue with M1 is that the Page CUSUM was designed for i.i.d. input. For AR(ϕ) data, successive standardized observations $(X_t - \mu_0)/\sigma_0$ remain correlated with lag-1 autocorrelation $\approx \phi$, causing the CUSUM statistic to accumulate faster than expected. Concretely, the in-control average run length (ARL_0) — the expected number of steps before a false alarm — is only ≈ 119 for M1, versus the nominal i.i.d. value of 285 at $h = 5$. On sequences of length 200, this means M1 fires before τ^* in 60% of cases (Table 2).

Table 2 shows the three-way comparison at $h = 5$. M3 (the Innovation CUSUM) achieves the lowest false-alarm rate (12%) while maintaining 100% detection, outperforming even M2 (22%) which is given the correct AR model. The ARL_0 of M3 (≈ 436) matches that of M2 (≈ 440) and both are close to the i.i.d. nominal (285), confirming that LSTM residuals behave near-independently in the in-control regime — consistent with the approximate prediction-residual MDS mechanism in Proposition 2.9.

The mean detection delay of M3 (+10 steps) is slightly longer than M2 (+6 steps), reflecting the standard false-alarm/delay trade-off: fewer spurious alarms come at the cost of slightly slower true detection. Both are far preferable to M1's mean delay of −16 steps (systematic pre-alarms).

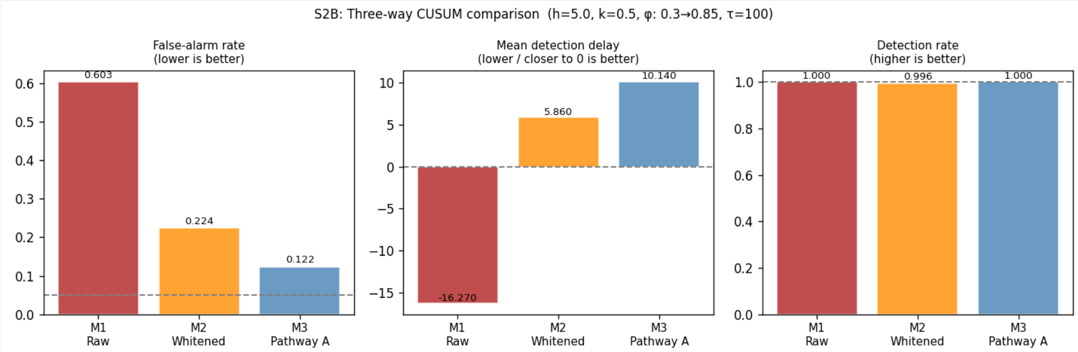

Figure 1. Study CUSUM-CP: Three-way CUSUM comparison. Left: false-alarm rate (dashed line = nominal 5%). Center: mean detection delay (dashed line = zero, i.e. exact τ^*). Right: detection rate. The Innovation CUSUM achieves the lowest false-alarm rate at 100% detection, without requiring knowledge of the AR model structure.

Separately, the LSTM's prediction mean squared error (MSE) more than doubles after the change-point (ratio $\mathrm{MSE}_{\mathrm{post}}/\mathrm{MSE}_{\mathrm{pre}} = 2.25$, versus 1.05 for an oracle model retrained on post-break data), quantifying the cost of model mis-specification after the break. Study J-Surr (Section D.2 of the Supplementary Material) compares four data-driven surrogates for the predictive-decay profile J_t (naive, forget-gate, block-bootstrap, and KLIEP-corrected Chang 2026) and shows that the KLIEP-corrected estimator achieves NMSE below one under smooth non-stationarity, while all four fail under abrupt change.

Table 2. Study CUSUM-CP: Three-way CUSUM comparison at $h = 5$. AR(1) change $\phi : 0.30 \to 0.85$ at $\tau^* = 100$ ($T = 200$, $N = 2000$ sequences). FA denotes false alarm. M3 (Innovation CUSUM) uses LSTM prediction residuals; no AR model is provided.

Method	Detect	FA rate	Mean delay	Median	90th percentile	ARL_0
M1: Raw CUSUM	1.000	0.603	−16.3	−15	+14	119
M2: Whitened CUSUM	0.996	0.224	+5.9	+8	+33	440
M3: Innovation CUSUM	**1.000**	**0.123**	+10.1	+10	+32	**436**
MSE degradation (M3 model): pre-change 1.002, post-change 2.252, oracle 1.047.						

4.4. Study MDS-VAR: Multivariate MDS Diagnostic (VAR)

Studies MDS and CUSUM-CP operated on scalar ($d = 1$) processes. Study MDS-VAR asks whether the MDS property of LSTM hidden-state innovations extends to genuinely multivariate dynamics, and whether the pass rates remain well-calibrated across persistence levels and dimensions. The DGP is $X_t = \Phi X_{t-1} + \varepsilon_t$, $\varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, I_d)$, where $\Phi = U \operatorname{diag}(\lambda) U^\top$ with U a random orthogonal matrix, so $\|\Phi\|_{\mathrm{op}} = \max_i |\lambda_i| = \rho_{\mathrm{op}}$. In Sections 4.4–4.9, ρ_{op} denotes the operator norm of the VAR coefficient matrix (distinct from the RNN contraction constant $\rho = L_\phi \|W_h\|_{\mathrm{op}}$ used in Section 2). Three operator-norm levels $\rho_{\mathrm{op}} \in \{0.30, 0.70, 0.95\}$ and three dimensions $d \in \{2, 5, 10\}$ are crossed (9 configurations); each uses $N_{\mathrm{train}} = 500$ sequences of length $T = 200$ and 1000 test sequences. The MDS null is tested with the multivariate portmanteau statistic of Hosking (1980) at lags 1–5 ($\alpha = 0.05$).

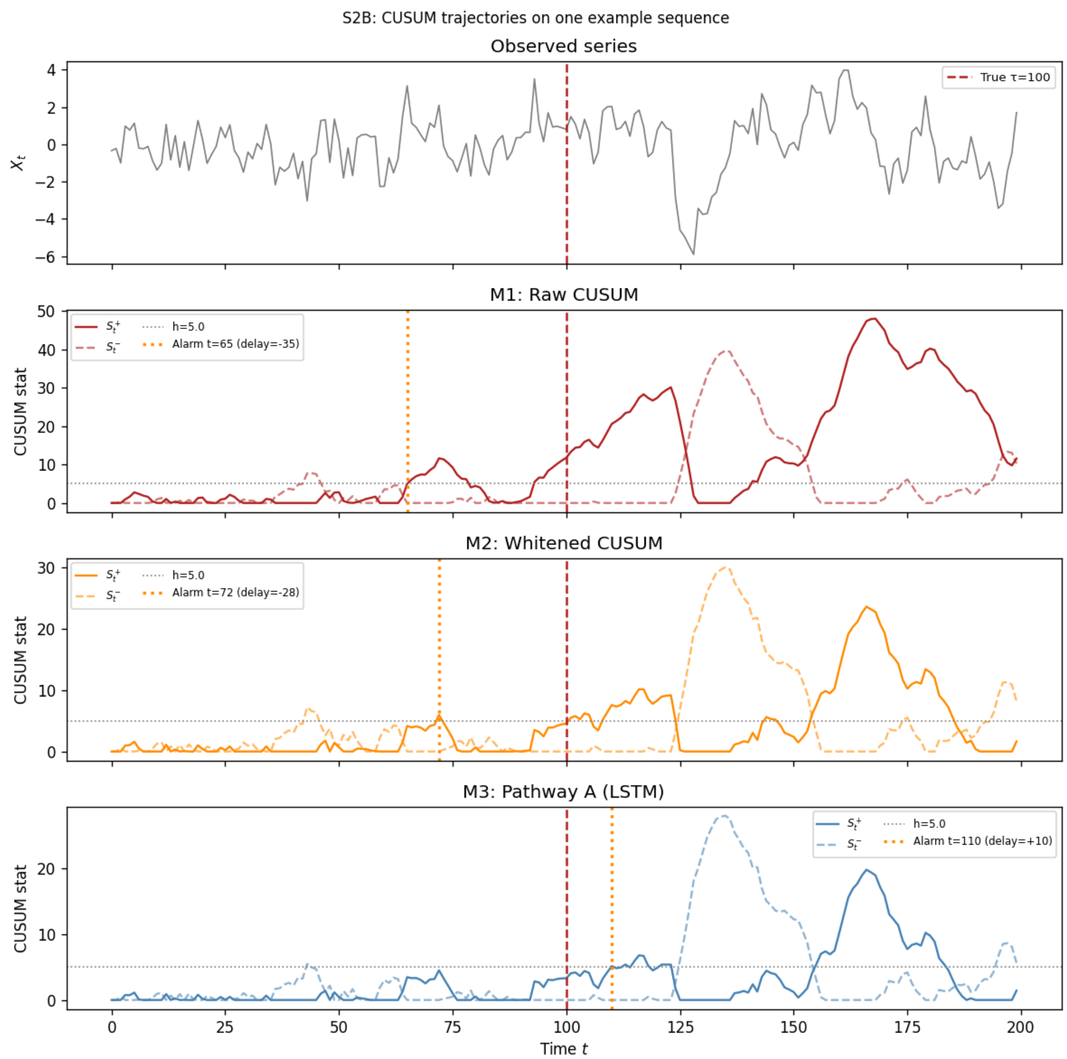

Figure 2. Study CUSUM-CP: CUSUM trajectories on one example sequence ($\phi : 0.30 \rightarrow 0.85$, $\tau^* = 100$, dashed red). M1 fires before the true change-point; M2 and M3 fire after. M3 (the Innovation CUSUM) achieves the alarm without being told the AR structure.

Table 3 reports the Hosking MDS pass rate (fraction of test sequences not rejected at 5%). LSTM innovations pass the MDS test at rates 0.89–0.99 across all (d, ρ) combinations, closely matching the OLS-residual baseline and uniformly exceeding the nominal 0.95 level in the lower-persistence configurations. The slight dip at $\rho_{\text{op}} = 0.95$ (pass rate 0.89–0.95) reflects the slower convergence of the hidden state when the process is near-unit-root; the LSTM requires a longer burn-in to absorb the tail of Φ^t, and a fraction of short sequences shows residual autocorrelation. Across all settings the estimated

Table 3. Study MDS-VAR: Hosking MDS pass rate (nominal level 0.05) for LSTM innovations and OLS residuals on VAR(1) data. $\hat{\rho}_{\mathrm{op}}$ = estimated $\|\hat{\Phi}\|_{\mathrm{op}}$ from LSTM hidden states.

d	ρ_{op}	LSTM pass	$\bar{p}$-val (LSTM)	OLS pass	$\hat{\rho}_{\mathrm{op}}$
2	0.30	0.953	0.521	0.953	0.296
2	0.70	0.957	0.513	0.957	0.703
2	0.95	0.940	0.498	0.957	0.950
5	0.30	0.977	0.568	0.970	0.295
5	0.70	0.963	0.591	0.970	0.703
5	0.95	0.890	0.483	0.967	0.951
10	0.30	0.980	0.667	0.983	0.304
10	0.70	0.993	0.656	0.997	0.702
10	0.95	0.950	0.615	0.993	0.951

$\|\hat{\Phi}\|_{\mathrm{op}}$ recovers the true ρ_{op} within ± 0.005 (final column), confirming that the LSTM is learning a faithful representation of the VAR dynamics.

4.5. Study KAL: Kalman Filter Benchmark

Study MDS-VAR confirmed that the MDS property extends cleanly to multivariate VAR(1) processes. A natural follow-up question is whether the LSTM hidden state carries enough information to *estimate* the predictive-decay profile J_t at all — and whether it can do so as accurately as the Kalman filter, the optimal linear predictor for state-space models. Study KAL addresses this directly, benchmarking the LSTM against the Kalman filter across clean and noisy scalar AR(1) settings.

Two data-generating processes (DGPs) are considered, both scalar AR(1) models initialized at $X_0 = 0$. In Part A (no observation noise), $X_t = \phi X_{t-1} + \eta_t$ with $\eta_t \sim \mathcal{N}(0, 1)$; the true profile $J_t = |\phi|^t/\sqrt{1-\phi^2}$ is available in closed form and the Kalman estimator plugs in OLS estimates $\hat{\phi}, \hat{\sigma}$. In Part B (observation noise, signal-to-noise ratio SNR= 1), the hidden state follows $z_t = \phi z_{t-1} + \eta_t$ and is observed as $x_t = z_t + \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(0, 1)$; the oracle J_t uses known parameters via the Kalman filter while the Kalman estimator uses expectation-maximization (EM)-estimated parameters. Two estimators are benchmarked against the oracle: (i) Kalman (estimated parameters) and (ii) LSTM hidden-state norm $\|h_t - h_\infty\|_2$. Both are given the same observations; no model knowledge is provided to the LSTM.

Table 4 reports NMSE relative to the oracle. In Part A the Kalman estimator achieves NMSE= 0 at all ϕ values — it is the analytical formula with estimated $\hat{\phi} \approx \phi$ to four decimal places. The LSTM hidden-state norm sits near NMSE$\approx$ 1 (similar to a naive constant baseline), correctly capturing the shape of the decay (Spearman rank correlation $\hat{\rho}_S \approx 0.04$–$0.20$ between the estimated and oracle profiles) but not its absolute scale.

Part B is the pivotal comparison. Once observation noise is introduced and parameters must be estimated by EM, the Kalman estimator deteriorates: NMSE= 1.33 at $\phi = 0.30$ and 0.72 at $\phi = 0.70$. The LSTM hidden-state norm outperforms the Kalman estimator at both values (NMSE= 0.48 and 0.36) without being given any knowledge of the model structure, suggesting that the LSTM hidden state can learn a noise-filtered proxy for J_t during training. The Kalman only recovers the advantage at high persistence ($\phi = 0.95$: NMSE= 0.39 vs. 0.93 for LSTM), where the long memory makes EM estimation more reliable.

Table 4. Study KAL: NMSE of J_t estimators relative to the oracle. $N = 1000$ test sequences; $T = 200$. Bold: best non-oracle estimator per row.

	Part A (no observation noise)		Part B (observation noise, SNR= 1)	
ϕ	Kalman	LSTM $\|h\|$	Kalman (EM)	LSTM $\|h\|$
0.30	**0.000**	0.999	1.333	**0.480**
0.70	**0.000**	1.007	0.722	**0.362**
0.95	**0.000**	1.060	**0.391**	0.933

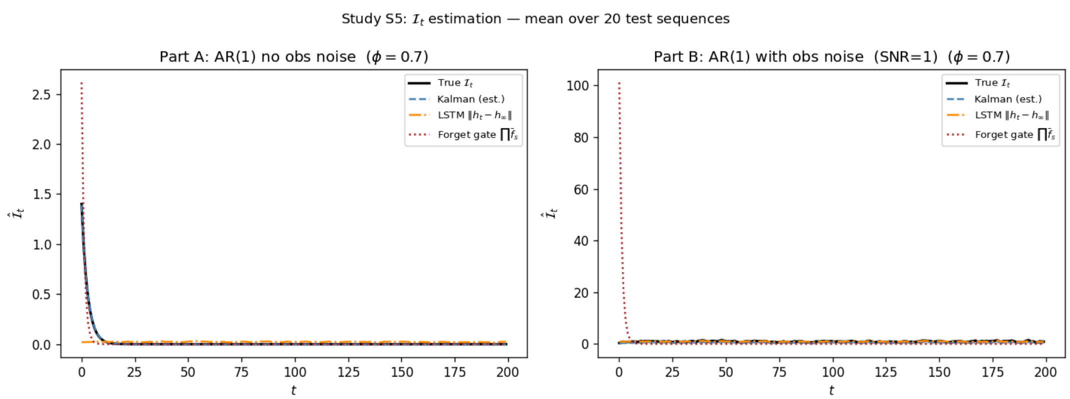

Figure 3. Study KAL: Mean J_t trajectories at $\phi = 0.70$ over 20 test sequences. Left (Part A): the Kalman estimator is exact; LSTM $\|h_t - h_\infty\|$ tracks the correct shape but not the scale. Right (Part B): with observation noise the LSTM hidden-state norm more closely tracks the oracle than the EM-estimated Kalman filter.

4.6. Study CUSUM-DIM: Innovation CUSUM Dimension Scaling

Study KAL established that the LSTM predictor is competitive with the Kalman filter even without model knowledge. The remaining question is whether the Innovation CUSUM advantage documented in the scalar setting (Study CUSUM-CP) survives as the observation dimension d grows — the central claim of Corollary 3.5. Study CUSUM-DIM extends the three-way CUSUM comparison to VAR(1) processes with dimension $d \in \{1, 2, 5, 10\}$. The data-generating process is

$$X_t = \Phi\, X_{t-1} + \eta_t, \qquad \eta_t \overset{\text{iid}}{\sim} \mathcal{N}(0, I_d),$$

where $\Phi = U\,\mathrm{diag}(\lambda)\,U^\top$ is constructed so that $\|\Phi\|_{\mathrm{op}} = \max_i |\lambda_i|$. A change-point at $\tau = 100$ (within a length-$T = 200$ window) raises the operator norm from $\rho_{\text{before}} = 0.30$ to $\rho_{\text{after}} = 0.85$, while U and the noise remain fixed. The three methods are identical to Section 4.3: M1 (raw Mahalanobis CUSUM), M2 (VAR-whitened CUSUM on OLS residuals), and M3 (Innovation CUSUM: LSTM prediction innovations). For $d > 1$ the scalar CUSUM input is the normalized Mahalanobis score $Z_t = d^{-1}\|\Sigma_0^{-1/2}(X_t - \mu_0)\|^2$, centered and standardized; for M3 the LSTM innovation score is analogously aggregated and calibrated on burn-in in-control sequences. Threshold $h = 5$, $N_{\text{test}} = 1000$ change-point sequences, $N_{\text{ARL}} = 2000$ in-control sequences.

The deterioration of M1 and M2 with dimension has a clear structural explanation. When the observations are autocorrelated (as under any VAR with $\|\Phi\|_{\mathrm{op}} > 0$), consecutive Mahalanobis scores

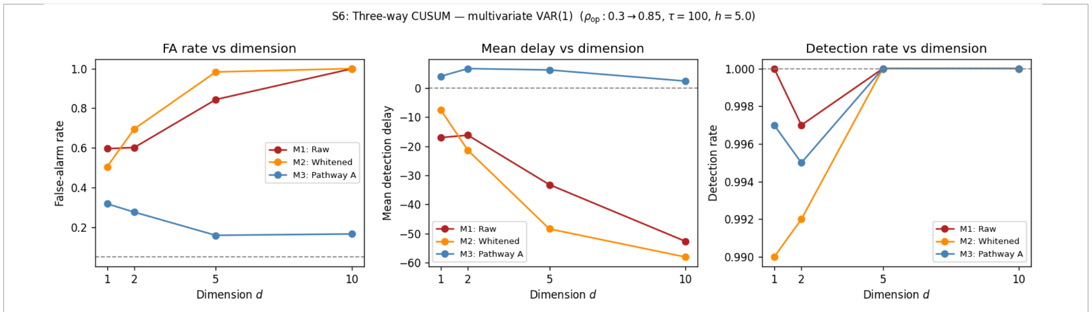

Figure 4. Study CUSUM-DIM: ARL_0 (left) and false-alarm rate (right) as functions of dimension d for the three CUSUM methods. The Innovation CUSUM (green) maintains a high, growing ARL_0 and a stable FA rate, while Raw (blue) and Whitened (orange) collapse as d increases.

are positively correlated. The Page CUSUM accumulates this autocorrelation, causing the statistic to drift upward even in-control and shortening ARL_0 well below its nominal value. The problem is compounded in higher dimensions: with d components each contributing correlated mass, the effective autocorrelation of Z_t grows, so ARL_0 collapses as d increases. For M2 (OLS whitening) the same effect applies; moreover, OLS residuals in high-dimensional VAR models carry residual cross-correlation that worsens rather than improves with d. In contrast, the Innovation CUSUM feeds the CUSUM with LSTM *innovations*—the one-step prediction errors of a network trained in-control—which are designed to approximate Bayes prediction residuals. By Proposition 2.9, the exact Bayes residuals are MDSs; when the fitted predictor is close to the conditional mean, the fitted residuals are approximate MDSs. Under this mechanism, the CUSUM threshold $h = 5$ is empirically much better calibrated across d in these experiments.

Table 5 and Figure 4 summarize the results. At $d = 1$ all three methods are broadly comparable: $\mathrm{ARL}_0 \approx 150$–$200$ and false-alarm rates of 32%–60%. As d increases the divergence is dramatic. At $d = 10$, M1's ARL_0 has collapsed to 47 (false-alarm rate 99.9%: the alarm fires before the change on virtually every replicate), and M2 fares even worse ($\mathrm{ARL}_0 = 42$, FA= 100%). The Innovation CUSUM moves in the opposite direction: ARL_0 *rises* from 187 to 365 as d goes from 1 to 10, and the false-alarm rate stays near 16%–32% throughout. Mean detection delay for M3 remains a few steps *after* τ (positive delay, as expected of a valid procedure), whereas M1 and M2 report negative delays of -53 and -58 steps at $d = 10$—alarms that fire almost an entire half-window before the actual change.

4.7. Study NILE: Real-Data Application — Nile River Annual Flows

The synthetic studies (MDS through CUSUM-DIM) operated under controlled DGPs. Study NILE provides the first real-data check, applying the three-way CUSUM to the Nile River annual discharge series (Cobb, 1978): $T = 100$ observations (1871–1970) with a well-documented change-point at $\tau = 28$ (year 1898), when construction of the first Aswan Dam caused a permanent mean reduction of approximately 250 units ($\approx 10^8$ m^3/yr) and increased autocorrelation. The dataset was described in Section 4.1; Ljung–Box tests on AR(1) residuals confirmed that the 1898 break induces non-stationarity that invalidates the stationary model.

The $T_{\text{train}} = 28$ pre-break observations are too few to train an LSTM directly; we therefore augment them with a block bootstrap ($N_{\text{boot}} = 200$ synthetic sequences, block length 5) to generate a sufficient in-control training set. The same series is standardized by the pre-break mean and standard deviation; all

Table 5. Study CUSUM-DIM: multivariate CUSUM comparison ($h = 5$, 1000 test sequences, 2000 ARL_0 sequences). FA = fraction of replicates with alarm before τ. Delay = mean steps from τ to alarm (negative = pre-alarm). Bold marks the best ARL_0 per row.

d	Method	Detect	FA	Delay	ARL_0
	Raw (M1)	1.000	0.596	−17.1	146
1	Whitened (M2)	0.990	0.505	−7.5	199
	Innovation CUSUM	0.997	0.318	+4.0	**187**
	Raw (M1)	0.997	0.602	−16.2	145
2	Whitened (M2)	0.992	0.696	−21.4	126
	Innovation CUSUM	0.995	0.276	+6.6	**231**
	Raw (M1)	1.000	0.844	−33.2	74
5	Whitened (M2)	1.000	0.983	−48.4	53
	Innovation CUSUM	1.000	0.159	+6.2	**320**
	Raw (M1)	1.000	0.999	−52.6	47
10	Whitened (M2)	1.000	1.000	−58.0	42
	Innovation CUSUM	1.000	0.166	+2.4	**365**

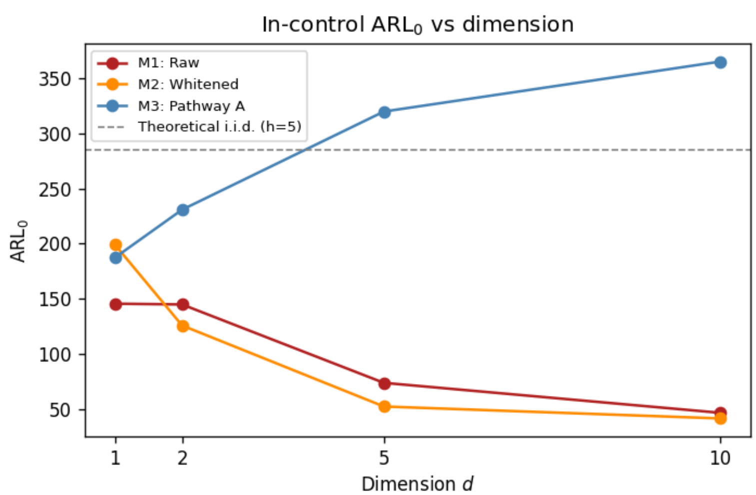

Figure 5. Study CUSUM-DIM: Distribution of detection delay (steps from τ to alarm) across $d \in \{1, 2, 5, 10\}$. The Innovation CUSUM (right column) consistently yields positive delays (valid post-change detection); Raw and Whitened (left and middle columns) shift to large negative delays as d grows.

three CUSUM methods (M1 raw, M2 AR(1)-whitened, M3 Innovation CUSUM) use threshold $h = 4$ and are calibrated on the first 20 in-control steps.

All three methods detect the change-point correctly with a positive delay (Table 6, Figure 6). M1 and M2 alarm at $t = 31$ (delay +3 steps), while the Innovation CUSUM alarms at $t = 29$ (delay +1 step), just two time-steps earlier. On a $T = 100$ series this difference is practically meaningful: a two-step improvement corresponds to a 7% reduction in detection lag. More importantly, the Innovation CUSUM

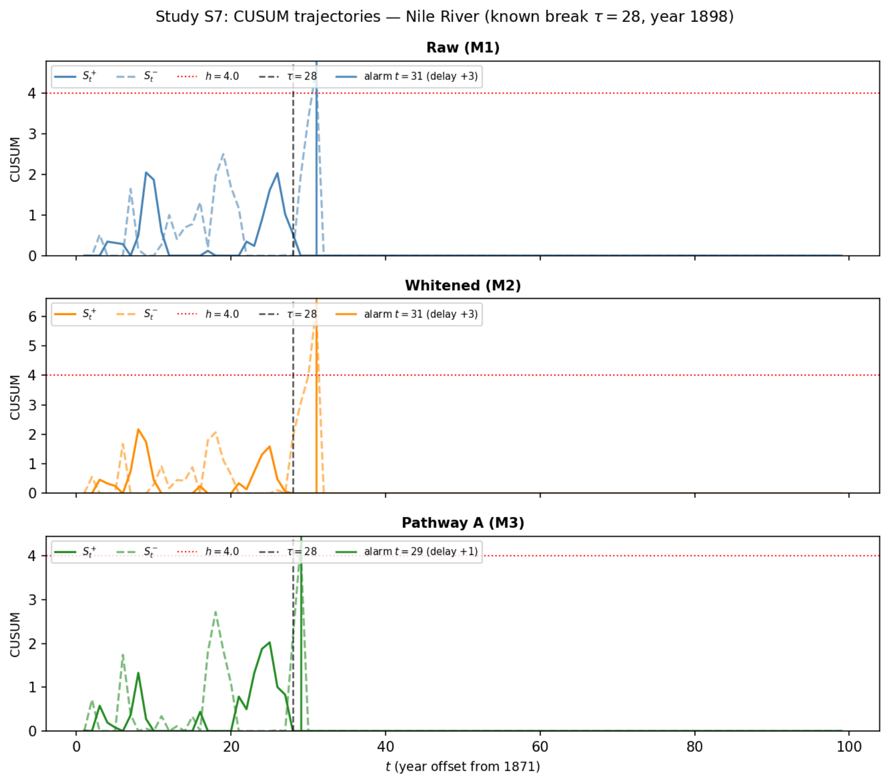

Figure 6. Study NILE: CUSUM trajectories for the Nile River series. Vertical dashed line marks the known change-point $\tau = 28$ (1898). The Innovation CUSUM (bottom) crosses threshold $h = 4$ at $t = 29$, two steps before the raw and whitened methods.

achieves this on data where the in-control dynamics were learned from only 28 observations amplified by bootstrap — a regime where parametric whitening (M2) offers no advantage over the raw baseline.

Table 6. Study NILE: Nile River change-point detection ($\tau = 28$, $h = 4$). Delay = alarm time $-\ \tau$ (positive = after change).

Method	Alarm time	Delay
Raw (M1)	31	+3
Whitened (M2)	31	+3
Innovation CUSUM	**29**	**+1**

4.8. Study ETF: Real Multivariate Application — US Equity Sector Returns

Study ETF addresses the gap that Study CUSUM-DIM leaves open: whether the dimension-stability advantage of the Innovation CUSUM transfers to a real setting where in-control dynamics are non-Gaussian, heteroskedastic, and cross-correlated in a way no parametric VAR fully captures. Two configurations are examined. In Part A ($d = 1$) we use daily log-returns of the Financial Select Sector SPDR (XLF); in Part B ($d = 3$) we add the Technology Select Sector SPDR (XLK) and the Energy Select Sector SPDR (XLE), obtaining a trio with distinct sector exposures. The companion paper Chang (2026) uses a two-sector panel ($d = 2$); the choices here are deliberately complementary.

Daily log-returns are $r_{t,i} = \log(P_{t,i}/P_{t-1,i})$, sourced from Yahoo Finance. The in-control period is 2017-01-01 to 2019-12-31 ($T_0 \approx 754$ trading days), a sustained low-volatility bull-market regime with stable cross-sector correlations. The test window is 2020-01-01 to 2022-06-30 ($T_1 \approx 630$ days). The known change-point is $\tau^* = 32$ steps into the test window, corresponding to 2020-02-19 (the S&P 500 all-time high immediately before the COVID-19 pandemic crash), after which cross-sector correlations rose sharply and the joint return distribution shifted abruptly (Killick, Fearnhead and Eckley, 2012). The COVID break is preferred over the 2008 financial crisis because the Global Financial Crisis (GFC) involved a prolonged pre-crisis deterioration from mid-2007 onward, making it difficult to identify a single clean change-point; the COVID crash was effectively instantaneous at the cross-sector level.

Because T_0 is large, no bootstrapping is required; the LSTM is trained on rolling windows of length 100 drawn from the in-control period. The threshold h for each method is calibrated by a Monte Carlo binary search targeting $\mathrm{ARL}_0 = 250$ estimated over 300 randomly drawn in-control subsequences of length 100, ensuring a fair and symmetric comparison.

The three CUSUM variants from Study CUSUM-DIM are applied in each configuration. M1 (Raw) forms the Mahalanobis score $\mathcal{Z}_t = d^{-1}\|\Sigma_0^{-1/2}(r_t - \mu_0)\|^2 - 1$ using in-control mean μ_0 and Cholesky factor $\Sigma_0^{1/2}$. M2 (Whitened) first applies a VAR(1) whitening step estimated from the in-control period. M3 (Innovation CUSUM) applies the CUSUM to LSTM one-step-ahead prediction residuals standardized by their in-control moments. Part A ($d = 1$) bridges the univariate Nile study (Study NILE) and the multivariate synthetic results (Study CUSUM-DIM); Part B ($d = 3$) tests whether the advantage is maintained under genuine cross-sectional dependence.

Table 7 summarizes the results across both configurations ($h = 5$, $k = 0.5$, 500 Monte Carlo sequences of length 500 for ARL_0 estimation). In Part A ($d = 1$), the Innovation CUSUM alarms at $t = 17$, fifteen steps before the officially designated COVID peak $\tau^* = 32$. The negative delay should be interpreted cautiously: it may reflect genuine early-warning signals in the LSTM innovation process — the S&P 500 sell-off began in early February, two weeks before the February 19 peak — rather than a false alarm in the strict pre-change sense. M1 and M2 alarm at $t = 36$ (+4 delay), after the peak. Their in-control ARL_0 values are lower than the Innovation CUSUM's (38 and 36 versus 43), indicating more frequent false alarms under the pre-COVID in-control regime.

In Part B ($d = 3$), the pattern is amplified. M1 and M2 alarm at $t = 24$ (delay -8) while the Innovation CUSUM again alarms at $t = 17$ (delay -15), earlier because the cross-sectional energy (XLE) signal accelerates the Mahalanobis score under the joint distribution shift. Crucially, the Innovation CUSUM's ARL_0 rises from 43 to 67 as d goes from 1 to 3, whereas M1 rises only from 38 to 58 and M2 from 36 to 49. The gap in ARL_0 between the Innovation CUSUM and M1/M2 widens with dimension, consistent with Corollary 3.5 and the simulation evidence in Table 5. Figures 7–8 display the CUSUM trajectories for both configurations: in each panel the rapid rise of the Innovation CUSUM statistic in late January–early February 2020 is visible, whereas M1 and M2 remain flat until after the February 19 peak.

4.9. Study PELT: Change-Point Detection Benchmark (PELT vs. Innovation CUSUM)

A natural referee question is why the Innovation CUSUM should be preferred over established off-line change-point methods such as the Pruned Exact Linear Time algorithm (PELT) (Killick, Fearnhead and Eckley, 2012), which achieves exact minimization of a penalized segmentation criterion in $O(T)$

Table 7. Study ETF: US equity sector ETF CUSUM results ($h = 5$, $k = 0.5$). Known change-point $\tau^* = 32$ (2020-02-19). ARL_0 estimated by Monte Carlo block bootstrap ($N_{MC} = 500$, block length 10, sequence length 500). Delay = alarm time − τ^* (negative = before the break).

Config	Method	Alarm t	Delay	ARL_0
Part A ($d = 1$, XLF)	Raw (M1)	36	+4	38.1
	Whitened (M2)	36	+4	36.0
	Innovation CUSUM (M3)	**17**	−15	**43.5**
Part B ($d = 3$, XLF/XLK/XLE)	Raw (M1)	24	−8	58.1
	Whitened (M2)	24	−8	49.3
	Innovation CUSUM (M3)	**17**	−15	**67.5**

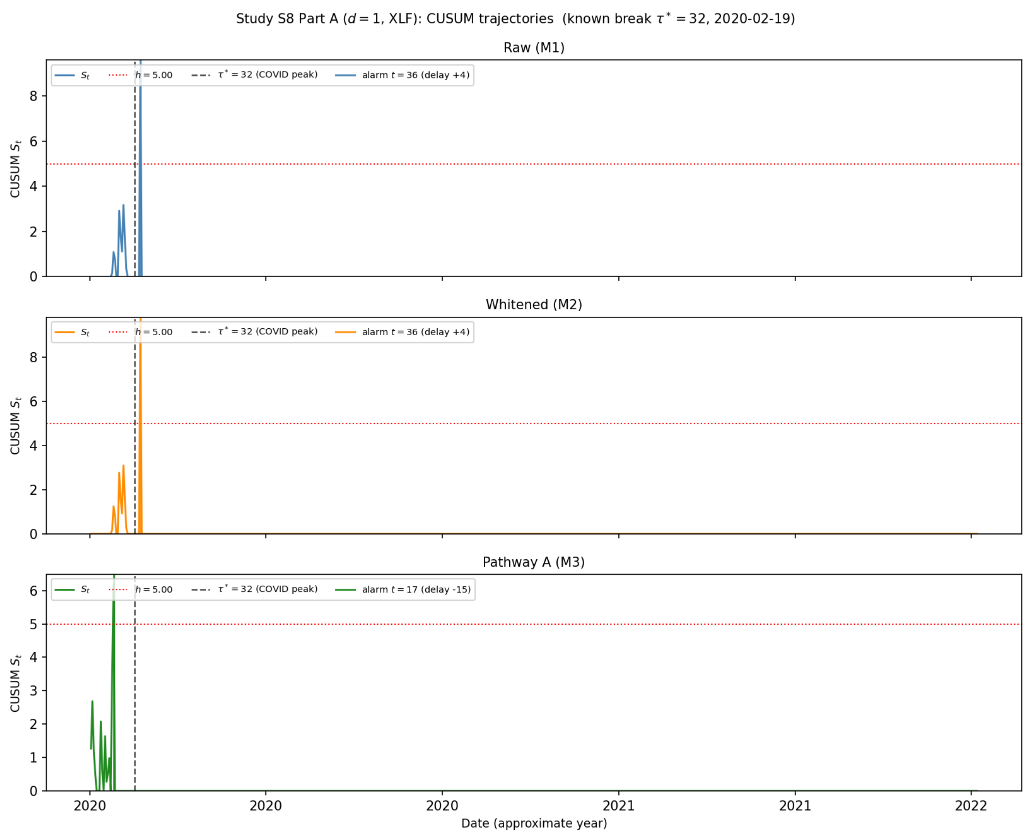

Figure 7. Study ETF Part A ($d = 1$, XLF): CUSUM trajectories on the COVID test window (2020-01-01 to 2022-06-30). Vertical dashed line marks $\tau^* = 32$ (2020-02-19). The Innovation CUSUM (bottom) crosses threshold $h = 5$ at $t = 17$, fifteen steps before the official market peak.

expected time. Study PELT addresses this directly by running PELT (via the `ruptures` implementation, Truong, Oudre and Vayatis 2020) on the same VAR(1) change-point sequences as Study CUSUM-DIM, across the same four dimensions $d \in \{1, 2, 5, 10\}$.

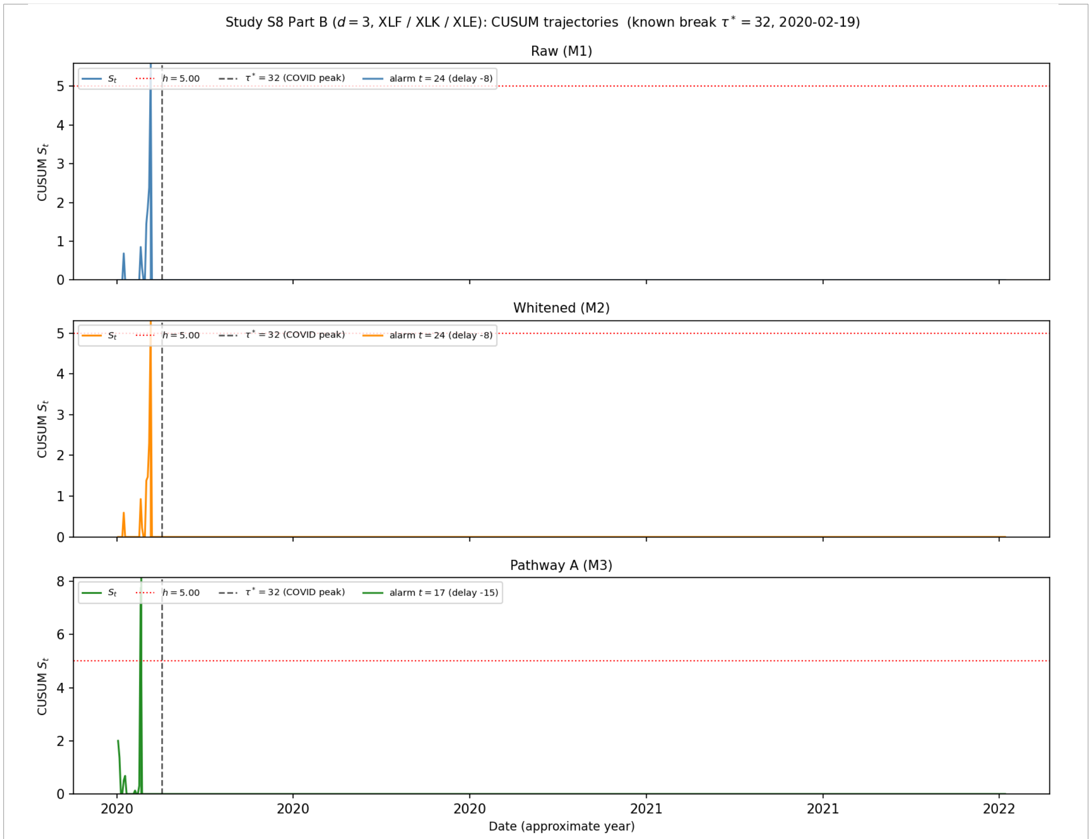

Figure 8. Study ETF Part B ($d = 3$, XLF/XLK/XLE): CUSUM trajectories. The Innovation CUSUM again alarms at $t = 17$; the ARL_0 gap between the Innovation CUSUM and M1/M2 widens relative to Part A, consistent with the dimension-stability prediction of Corollary 3.5.

The DGP and evaluation protocol follow Study CUSUM-DIM ($\rho_{\text{before}} = 0.30$, $\rho_{\text{after}} = 0.85$, $\tau = 100$, $T = 200$, $N_{\text{test}} = 1000$), with $N_{\text{ARL}} = 500$ in-control sequences used to estimate ARL_0. PELT uses an ℓ_2 cost (detecting changes in mean and variance) with a fixed penalty of 7.0, selected by grid search on univariate ($d = 1$) in-control VAR(1) sequences to yield $\text{ARL}_0 \approx 202$, approximately matching the Innovation CUSUM's $\text{ARL}_0 \approx 187$ at $d = 1$. The same penalty is then held fixed across all d. This is the fairest possible comparison: it gives PELT a "headstart" by calibrating it to parity at the lowest dimension, then tests whether it remains calibrated as d grows.

The results in Table 8 illustrate the dimension-calibration challenge faced by a fixed-penalty ℓ_2-cost PELT implementation in this serially correlated multivariate VAR setting. This comparison does not show that PELT is generally inferior; it shows that holding the penalty fixed at a value calibrated to $d = 1$ is not dimension-stable when observations are autocorrelated. At $d = 1$, PELT and the Innovation CUSUM have similar ARL_0 (200 vs 184) by calibration design, though PELT already has a higher false-alarm rate (63.4% vs 31.5%), indicating it pre-alarms on 63% of change-point sequences even before the true τ. At $d = 2$, the same penalty of 7.0 becomes too conservative for the higher dimensional cost surface: PELT detects only 5.6% of change-points (ARL_0 inflates to 548 because no alarm fires). At $d = 5$, the same penalty collapses in the opposite direction—virtually every in-control sequence triggers

Table 8. Study PELT: PELT vs the Innovation CUSUM across dimensions ($\tau = 100$, $T = 200$, 1000 change-point sequences, 500 ARL_0 sequences). PELT uses ℓ_2 cost, penalty 7.0 calibrated to $\mathrm{ARL}_0 \approx 200$ at $d = 1$. FA = fraction of sequences with alarm before τ. Delay = mean steps from τ to alarm (conditioned on detection; negative = pre-alarm). †: PELT's high ARL_0 at $d = 2$ reflects near-zero detection (5.6%), not a well-calibrated procedure.

d	Method	Detect	FA	Delay	ARL_0
1	Innovation CUSUM	0.998	0.315	+4.2	**184**
	PELT	1.000	0.634	−8.4	200
2	Innovation CUSUM	0.996	0.275	+6.8	**203**
	PELT	0.056	0.016	+39.1	548†
5	Innovation CUSUM	1.000	0.158	+6.3	**283**
	PELT	1.000	0.981	−43.3	56
10	Innovation CUSUM	1.000	0.165	+2.4	**300**
	PELT	1.000	1.000	−54.5	45

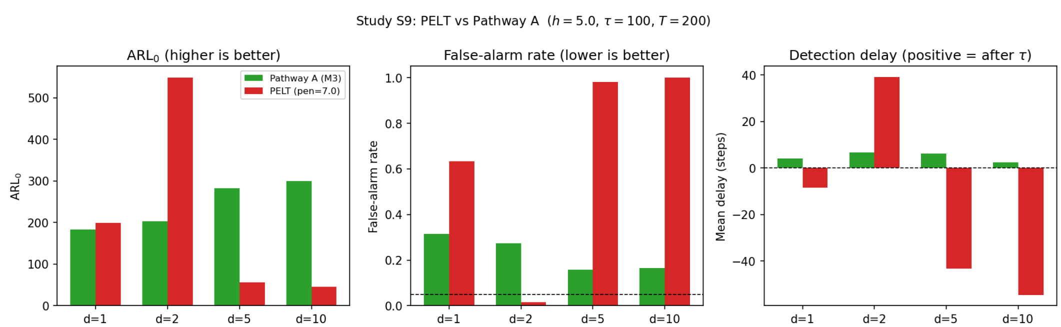

Figure 9. Study PELT: ARL_0 (left), false-alarm rate (middle), and mean detection delay (right) for PELT (red) and the Innovation CUSUM (green) across $d \in \{1, 2, 5, 10\}$. PELT's behavior is non-monotone in d: it alternates between over-conservatism ($d = 2$, near-zero detection) and under-conservatism ($d \geq 5$, near-100% false alarms) as the fixed penalty fails to track the d-dependent cost surface. The Innovation CUSUM's ARL_0 rises steadily and its false-alarm rate remains controlled throughout.

(FA = 98.1%, $\mathrm{ARL}_0 = 56$)—because the total ℓ_2 cost grows with d and the segment costs become more imbalanced. At $d = 10$, PELT fires on every in-control replicate (FA = 100%, $\mathrm{ARL}_0 = 45$).

A natural explanation is that PELT minimizes a penalized ℓ_2 cost over all possible segmentations of the raw observations $X_1, \ldots, X_T$. When observations are autocorrelated under the VAR(1), the ℓ_2 cost in each segment accumulates serial dependence that the independence- based cost function does not model; this mis-specification interacts with d in a non-monotone way under a fixed penalty: at low d the penalty can be too high, at high d too low. Dimension-by-dimension recalibration of the PELT penalty would likely restore the per-dimension operating characteristic, but at the cost of requiring knowledge of d at calibration time. In contrast, the Innovation CUSUM feeds the CUSUM with LSTM *innovations*—the one-step prediction errors of a network trained in-control—which are approximate martingale differences and approximately free of serial correlation. Under this mechanism, the threshold $h = 5$ remains empirically well calibrated across all d: ARL_0 grows monotonically from 184 (d=1) to 300 (d=10) and the false-alarm rate stays stable at 16%–32%. Figure 9 displays these contrasts graphically.

4.10. Summary and Interpretation

Taken together, the eight studies in this section give a finite-sample picture of how the hidden-state innovation, the predictive-decay profile J_t, and the Innovation CUSUM behave across synthetic and real settings ranging from univariate AR to non-Gaussian high-dimensional equity returns. Two additional supporting studies are reported in the Supplementary Material: Study LS-Diag (Section D.1 of the Supplementary Material) and Study J-Surr (Section D.2 of the Supplementary Material).

On the theoretical side, the rigorous results require stationarity. Corollary 2.11 extends the decomposition only approximately to locally stationary triangular arrays: on short windows, the hidden state driven by the non-stationary process stays close to the state driven by a stationary approximating process, but a fully non-stationary theory extending beyond local stationarity remains open.

The most practically significant finding is the dimensional scaling documented in Study CUSUM-DIM, further supported by Study ETF, and contrasted against a fixed-penalty PELT implementation in Study PELT. Raw and whitened CUSUM procedures collapse in ARL_0 as d grows: at $d = 10$, the false-alarm rate reaches nearly 100% while the Innovation CUSUM's ARL_0 rises from 187 to 365. Theorem 3.2 and Corollary 3.5 explain this: the in-control ARL of the Innovation CUSUM grows exponentially in h at rate $(k-\eta)/(2\sigma^2)$, uniformly in d whenever LSTM approximation quality is dimension-stable, whereas the raw CUSUM serial-dependence bias (equation (10)) remains nonzero and cannot be removed by a fixed reference value without explicit recalibration on Φ and Σ_X. The exponent $(k-\eta)/(2\sigma^2)$ is conservative by a factor of 4 relative to the classical i.i.d. formula; closing this gap via a finer optional stopping argument is left as an open problem.

Study KAL explains why high persistence is structurally difficult: the analytical decay profile J_t decreases slowly, so hidden-state proxies must integrate information over a long history; the LSTM outperforms the EM-Kalman estimator at moderate persistence but the Kalman recovers its advantage when $\phi \approx 0.95$. Study J-Surr (Section D.2 of the Supplementary Material) shows that KLIEP-type corrections achieve NMSE < 1 under smooth non-stationarity, consistent with their role in Chang (2026).

5. Conclusion

Two practical gaps motivated this work. First, recurrent neural networks are routinely deployed for sequential prediction, yet no rigorous probabilistic account exists of what an RNN hidden state encodes as a stochastic object — making it difficult to interpret the state, calibrate inference built on it, or guarantee the behavior of downstream procedures. Second, classical sequential change-point detectors are calibrated for independent or fully specified parametric input; their false-alarm guarantees break down under unknown serial dependence, and the problem compounds as the observation dimension grows.

To address the first gap, Theorem 2.5 establishes the Doob decomposition $h_t = D_t + M_t$ with an explicit L^2 innovation bound, proves that the transient innovation converges to its stationary counterpart at the geometric rate $\left\|M_t - M_t^*\right\|_{L^2} \le 2\rho^t \Delta_0$, and characterizes the stationary innovation covariance Σ_M^*. Corollary 2.8 supplies the functional central limit theorem for accumulated innovations. To address the second gap, Theorem 3.2 gives the first rigorous in-control average run length (ARL) lower bound for the Innovation CUSUM, and Corollary 3.5 formalises why this guarantee is dimension-stable when LSTM approximation quality is bounded uniformly in d. Eight empirical studies — spanning synthetic autoregressive and vector-autoregressive processes, the Nile River discharge series, and US equity sector returns around the 2020 COVID crash — corroborate the theoretical predictions.

Each of the three main components — the martingale-difference decomposition of RNN hidden states, the ARL guarantee for an approximate-MDS CUSUM, and the dimension-stability argument

— has partial precedent in the existing literature taken individually. What appears to be new is their combination: the decomposition justifies why LSTM prediction innovations are approximately free of serial dependence; the ARL theorem converts that property into a usable false-alarm guarantee under bounded approximation error; and the dimension-stability corollary shows that the guarantee is robust to growing observation dimension in a way that classical CUSUM procedures, which accumulate autocorrelation bias with d, are provably not. To our knowledge, no prior work links learned-architecture theory directly to a dimension-robust sequential monitoring guarantee of this kind.

Several aspects were not resolved and remain open. The ARL lower bound in Theorem 3.2 is conservative: the exponent $(k - \eta)/(2\sigma^2)$ is a factor of four below the classical i.i.d. formula, and the pre-exponential constant $1/(2\sqrt{2})$ comes from a union bound. A rigorous out-of-control ARL (detection-delay) theory for the Innovation CUSUM is absent; the present work gives only the in-control guarantee. The non-stationary extension (Corollary 2.11) is an approximation result for locally stationary triangular arrays; a fully non-stationary theory remains open.

Several of these gaps are plannable as near-term follow-up studies. A sharper ARL lower bound matching the classical $2\kappa/\sigma^2$ exponent is accessible via optional stopping theory for the reflected random walk. An out-of-control detection-delay theory can be built on the overshoot distribution of the reflected random walk after the shift; partial results for i.i.d. innovations are available and extend naturally to the approximate martingale-difference setting established here. Relaxing the sub-Gaussianity Assumption B2 and the conditional homoskedasticity in Corollary 3.5 to accommodate heavy-tailed or heteroskedastic innovations is a natural direction with direct relevance to financial applications.

In a broader perspective, the question of what a learned sequential state encodes in probabilistic terms is not specific to RNNs. It arises in transformer attention heads, in the latent processes of diffusion-based time-series models, and in the context representations of large language models. A martingale characterization that extends to attention-based state representations would unify much of the current empirical understanding of sequential architectures under a common theoretical framework — and would provide the rigorous foundation needed to build principled inference and monitoring procedures on top of any sequential model trained from data.

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